The A-Z of Geo Skills

Part 2: N to Z

W

Weather (1)

The A-Z of Geo Skills

- Weather can be defined as the state of the atmosphere at a particular place & time
- Weather is different from climate which is the state of the atmosphere over a prolonged period of time
- Weather only happens in the troposphere which is the layer of the atmosphere closest to earth (16km high)
- Because the Earth's surface varies from hot to cold and sea to land the atmosphere above also varies – the weather we feel is caused by the movement of air masses

Temperature (°C)	Descriptive term
over 35	very hot
25 to 35	hot
15 to 25	cool/warm
5 to 15	cold
–5 to 5	very cold
below –5	freezing
Rainfall intensity (mm/h)	**Descriptive term**
0.0 to 2.5	very light
0.25 to 1.00	light
1.00 to 4.00	moderate
4.00 to 16.00	heavy
16.00 or more	very heavy

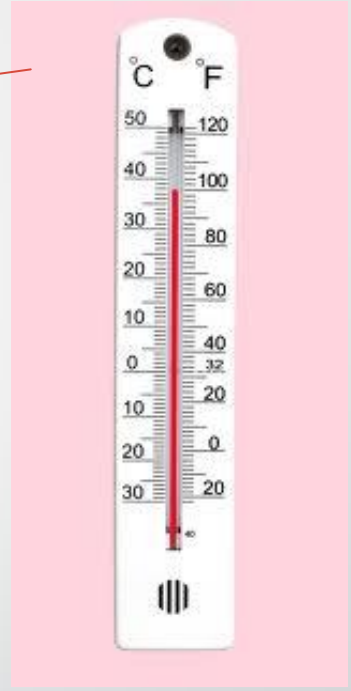
Descriptive terms for the weather

W

Weather (2)

The A-Z of Geo Skills

Element	Description	Measured by
Temperature	The warmth of the surrounding air caused by the Sun	Thermometer
Precipitation	All forms of moisture – rain, hail, sleet, snow, dew and frost	Rain gauge
Humidity	The amount of moisture the air can hold at a particular temperature	Hygrometer
Air pressure	The 'weight' of air pressing on the Earth's surface	Barometer
Sunshine	Hours of sunlight in a day	Sunshine recorder
Wind	The movement of air of varying warmth and moisture named by the direction from which it blows	Anemometer (km/hr) *also* Beaufort Scale
Cloud Cover	Concentrations of water vapour, named by their shape, size, height and thickness – measured in eighths with zero being clear	Human observation



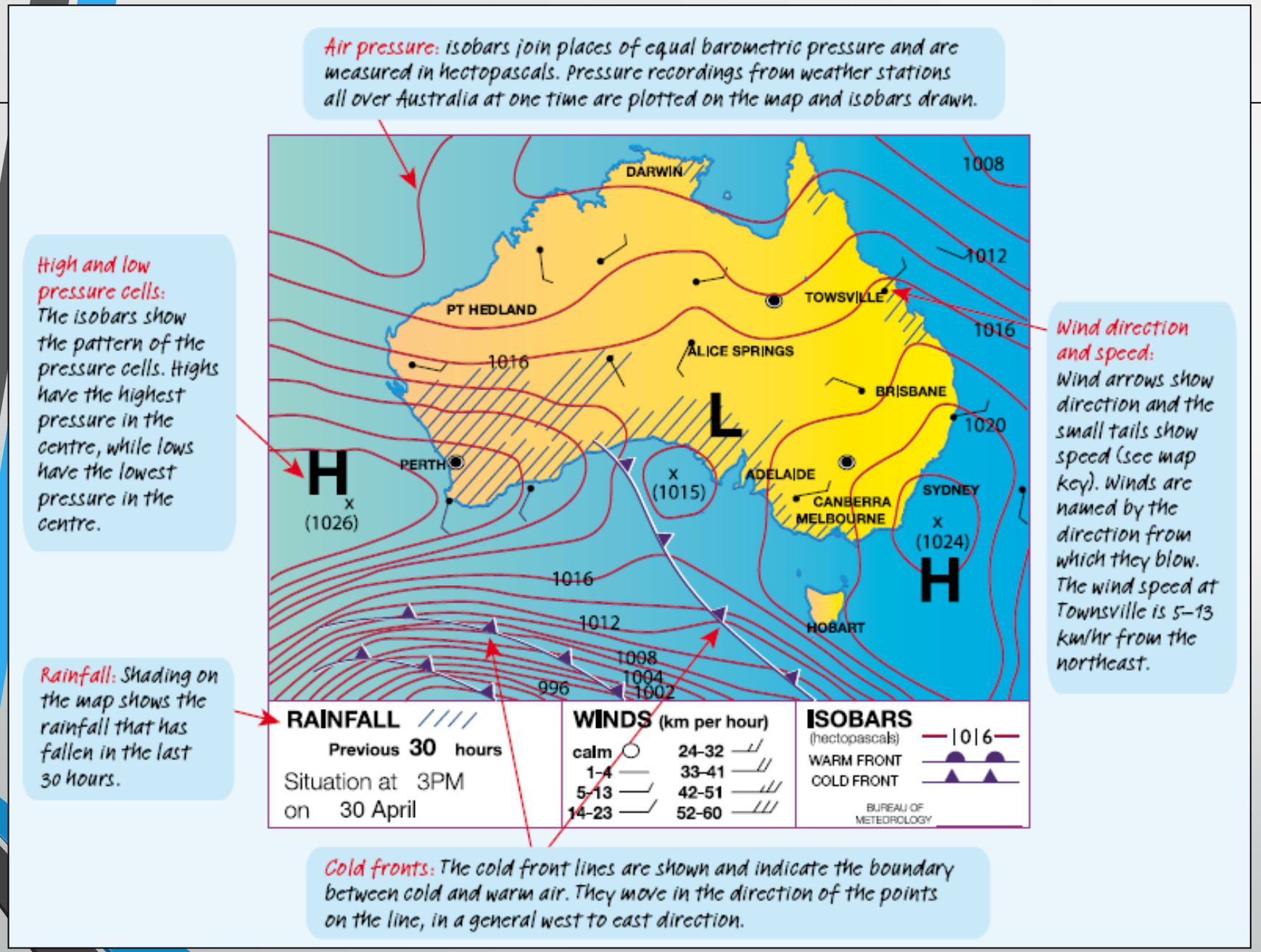






The elements of weather

Weather (3)



Reading a weather map (aka a synoptic chart)

- A weather map is also called a <u>synoptic chart</u>
- This shows us a synopsis of the weather conditions at a particular time
- The main features are: isobars, cells, fronts, winds and rainfall (shown adjacent)

Weather (4)

- Exam style questions

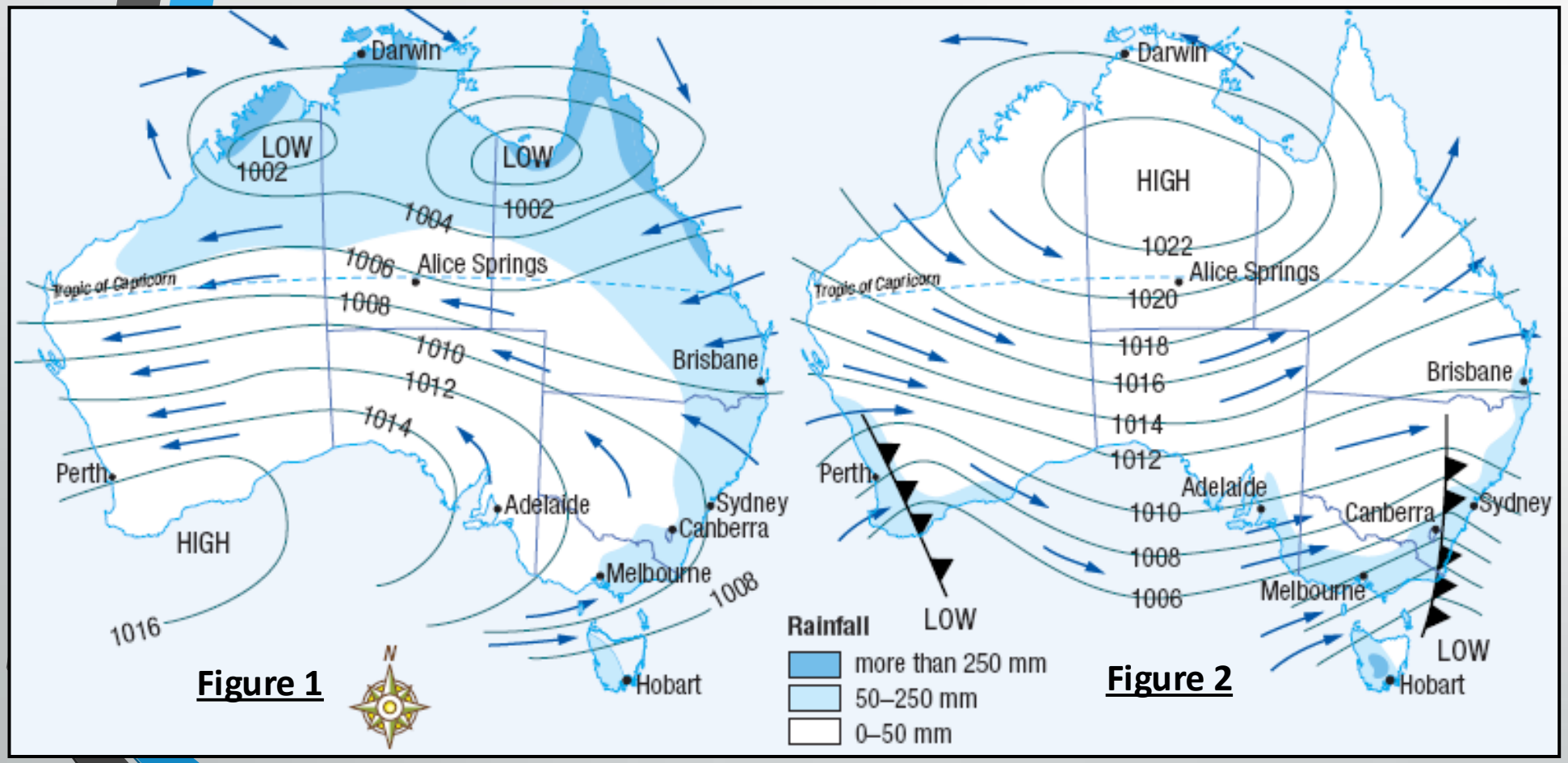


Figure 1

Figure 2

Exam style questions:

Q1) In Fig. 1, what is the air pressure in Adelaide?

Q2) In Fig. 2, what is the weather feature approaching Sydney?

Q3) In Fig. 1 how much rainfall is in Brisbane ?

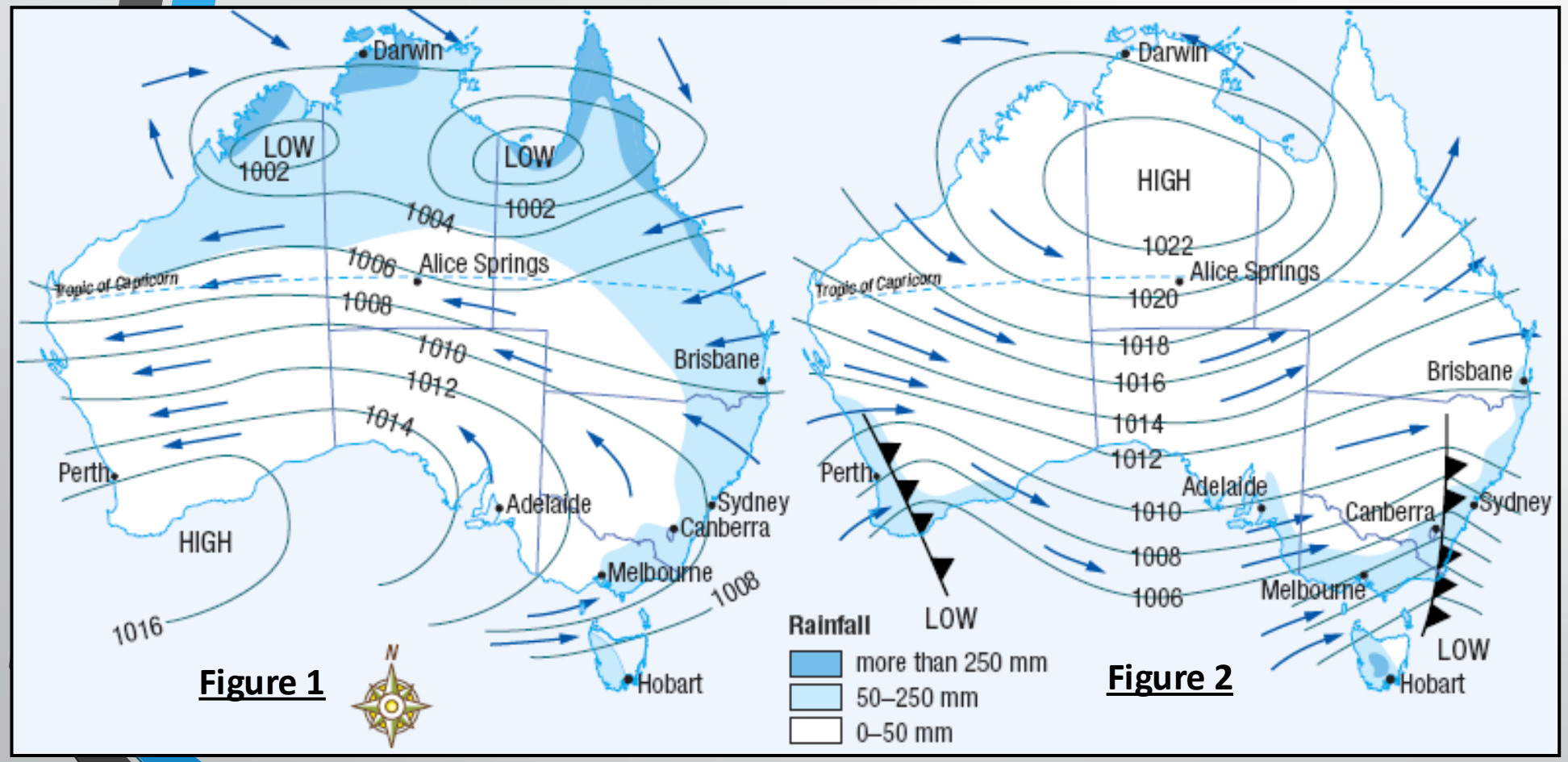
Q4) In Fig. 1 which season do you think is represented and why?

Q5) In Fig. 2 what is the wind direction at Perth?

Q6) In Fig. 2, describe the weather conditions in Alice Springs.

Weather (4)

- Exam style questions



ANSWERS:

Q1) In Fig. 1, what is the air pressure in Adelaide?
1012 hPa – 1014 hPa

Q2) In Fig. 2, what is the weather feature approaching Sydney?
Cold front

Q3) In Fig. 1 how much rainfall is in Brisbane ?
50-250 mm

Q4) In Fig. 1 which season do you think is represented and why?
Summer, wet season and low cells in the Nth

Q5) In Fig. 2 what is the wind direction at Perth?
SW

Q6) In Fig. 2, describe the weather conditions in Alice Springs.
Fine, sunny, settled, clear skies, light westerly breeze

The End

Phew…

Good luck in mastering your Geo Skills